

175 Days Lifestyle Challenge

Development & reflection phase

A tested Python tracker for daily and weekly progress

175 days makes progress ambitious yet finishable

The challenge is long enough for repeated daily and weekly actions to become visible, yet finite enough to feel approachable. It is a product milestone, not a scientific guarantee of behaviour change.

175

days

A clear finish line under six months

5

starting habits

Useful from the first launch

2

supported periods

Daily and weekly

Three simple actions carry the whole journey

START



1. Create

Define a clear task and choose a daily or weekly period.

EACH PERIOD



2. Check off

One completion counts in the active calendar period.

ANY TIME



3. Learn

Compare longest streaks and decide what to improve next.

Five starting habits make day one useful

1 WEEKLY

Write a weekly reflection

Review progress, record how the week felt, and choose one improvement for the next week.

The bundled fixture provides four completed weeks.

DAILY 01

Wake before 7 a.m.

DAILY 02

Learn a useful skill for one hour

DAILY 03

Read non-fiction for 15 minutes

DAILY 04

Exercise for 30 minutes

A layered design keeps every check-off valid and stored

Command-line interface

Turns add, list, check, delete and analytics commands into application calls. It contains no SQL.

Python application

Habit validates domain rules. The repository controls persistence. Pure functions calculate streaks.

SQLite database

Habit definitions and completion timestamps persist locally with indexed lookups and cascade deletion.



USER



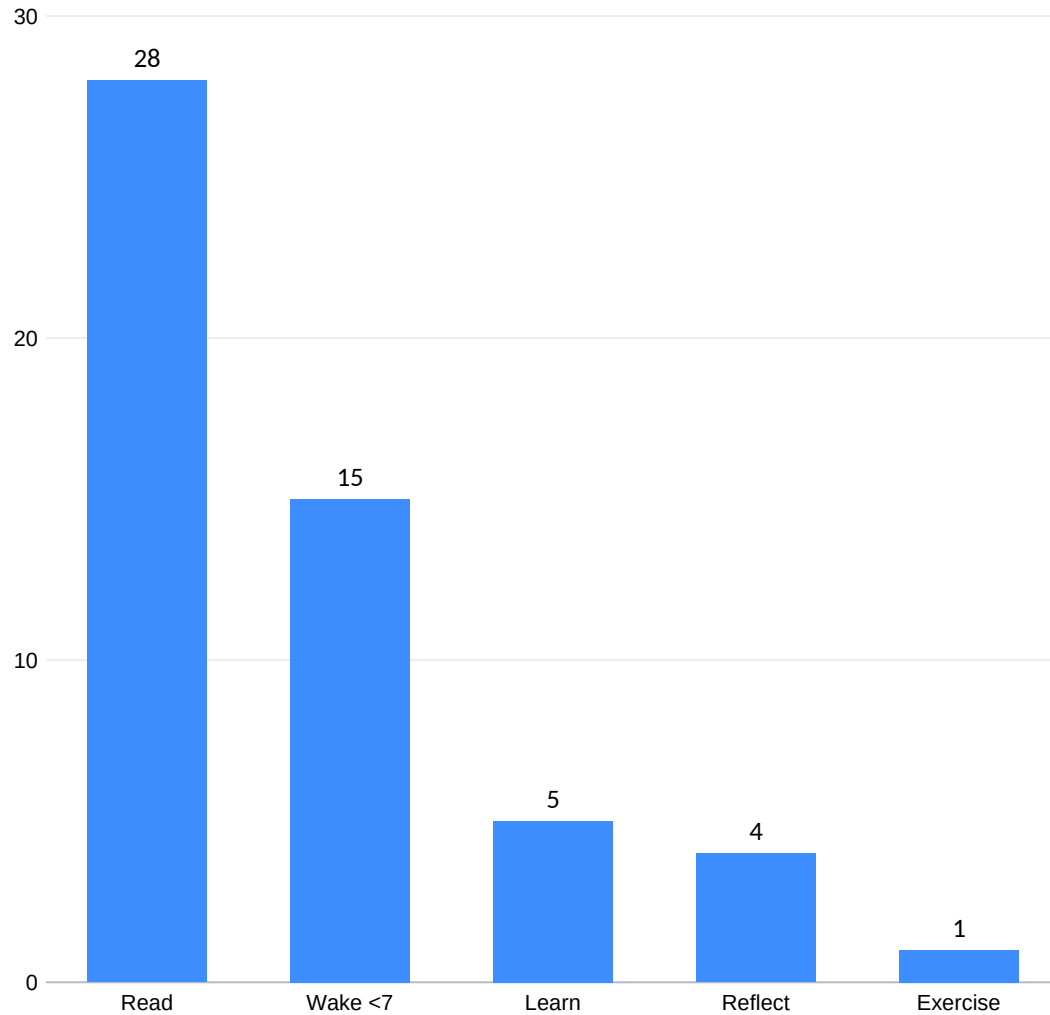
LOGIC



DATA

Fixture data proves the analytics

LONGEST STREAK · PERIODS



Pure functions

map, filter and reduce calculate results without database access or mutation.

Both periods

Daily dates and ISO weeks share the same consecutive-period rule.

Fixture result

Reading leads with 28 completed daily periods in a row.

Setup takes one command; daily use stays human-readable

```
$ python main.py init
Created 5 habits + 4 weeks of data

$ python main.py list --periodicity daily
4 daily habits

$ python main.py check 4
Checked off 'Exercise for 30 minutes'

$ python main.py analytics
Overall longest streak: 28 periods
```

Install

Python 3.7+ is enough; runtime dependencies are all standard library.

Track

Use a habit name or numeric ID. Duplicate check-offs in one period are rejected.

Verify

Run `python -m unittest -v`. All eight unit and integration tests pass.

Implementation feedback made the design stronger

Tutor feedback questioned why the challenge lasts 175 days and whether the interface should talk directly to the database. Implementation converted both questions into clearer product and architecture decisions.

CONCEPTION

Interface ↔ database

The theme was clear, but the technical boundary was too shallow.

IMPLEMENTED

CLI → Python core → SQLite

Repository-backed rules, pure analytics, persistent data and 8/8 passing tests.

The phase-2 prototype is working, documented and ready for tutor feedback.